CHAPTER

7

Asking Questions About Threatening Topics: A Selective Overview

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This view [that not all intentionally false statements ought to count as lies] found powerful expression in Grotius. He argued that a falsehood is a lie in the strict sense of the word only if it conflicts with a right of the person to whom it is addressed.

—Bok (1978, p. 37)

In their 1950 study that included a comparison between survey responses and records, Parry and Crossley (1950) showed that respondents overreported voting, having a library card, and several other socially desirable characteristics as compared to information about the respondents in external records. In a 1976 record-check study, Locander, Sudman, and Bradburn reported similar findings and also showed that respondents underreport bankruptcy and being charged with drunken driving. These classic studies addressed a key concern of those who rely on survey data—the accuracy of responses to questions that respondents might perceive as threatening.

All self-reports are subject to response error. But validation studies, which incorporate a comparison between self-reports and records, suggest that some respondents sometimes perceive some topics as threatening, so that reports about threatening topics are subject to more than the usual errors caused by omissions, telescoping, and the like. Results like those just mentioned suggest that asking a threatening question seems to elicit from some respondents a tendency to present themselves in a socially desirable way. This tendency may lead to overreports of socially desirable behaviors or to underreports of threatening behaviors.

Instead of beginning with a discussion of what makes a question sensitive in a way that is socially or interactionally threatening, I first present excerpts from research on three topics that could be considered threatening. The topics, which have all been the subject of recent studies, are the number of sex partners a person has had, whether a woman has ever had an abortion, and the use of marijuana. The discussion will illustrate why we think we know the direction of errors in self-reports on these topics and the range of techniques that have been used in an attempt to reduce the component of response error caused by the threatening nature of the topics. The techniques that have been used include varying the mode of asking and answering questions, varying the context of the questions within the interview, and varying the structure or wording of the questions. After presenting illustrative results, I will then consider what theoretical developments have attempted to integrate these findings, what makes a question threatening, and what other avenues might be investigated to improve the accuracy of self-reports on such topics.

REPORTING OF THE NUMBER OF SEX PARTNERS

Consider first reporting about the number of sex partners by heterosexuals. In a series of comparisons, Smith (1992) found that, on average, men report many more sex partners than do women. Table 7.1 illustrates his findings. The first row presents the unadjusted estimate from the 1989 General Social Survey; the second row presents the results with some reasonable adjustments made to include nonrespondents; and the third row adds an adjustment that reduces the impact of extreme values, which are more likely to be reported by men, on estimates of the means. The table shows that the difference is substantial and that it persists despite attempts to correct for

TABLE 7.1
Mean Number of Adult Lifetime Sex Partners
Reported by Heterosexual Men and Women

	Males	*Females*	*Males:Females*
Unadjusted	13.00	3.24	4.06:1
Adjusted for nonresponse*	12.05	3.03	3.98:1
Adjusted for extreme values**	9.36	3.02	3.10:1

Note. Male and female means different at < .01 in all rows.

Source: 1989 General Social Survey. From Table 7 in Smith (1992). Copyright © 1992 by Statistics Sweden. Reprinted by permission of the author and Statistics Sweden.

*Values of 1.0 given to males and females with missing data.

**Values of 50 and greater recoded to 50.

possible sources of the discrepancy other than response error. Smith reported finding the difference in other surveys, and the difference is large enough that adjustments for obvious possible sources of the discrepancy, such as differences between the men and the women who participate in surveys or in the way men and women use extreme values when answering, do not make men's and women's reports the same. This comparison involves a relatively closed population; thus, reports from men and from women should yield similar estimates of population means. The fact that they do not, even when attempts are made to adjust the data, suggests that the number of sex partners is being misreported. Smith (1992) argued that, "most probably there is a combination of male overreporting and female underreporting" (p. 320). Thus, methods that decrease the number of partners that men report or increase the number reported by women might be interpreted as improving reports.

A series of experiments tested several methods for improving reporting of sex partners. These experiments varied the mode of response, the structure of the question, and question context. The results presented here from the Women's Health Study are based on a sample of women selected by area probability methods in Chicago and women selected from two Chicago health clinics. A comparison of self-administered questions with interviewer-administered questions (Table 7.2), found that, overall, the mean number of sex partners reported by women was significantly higher when self-administered methods of answering questions were used, and the effect was significant for three reference periods: the past year, the past 5 years, and over the respondent's lifetime.

A later experiment (referred to by the investigators as the ACASI experiment) compared audio-enhanced, computer-assisted self-interviewing (ACASI),

TABLE 7.2
Mean Number of Sex Partners Reported by Women, by Mode of Interview

Mode of Interview (Experimental Group)	*Mean Reported Sex Partners*		
	Past Year	*Past 5 Years*	*Lifetime*
Self-administered questions (self-administered or CASI)	1.72	3.88	6.54
Interviewer-administered questions (paper or CAPI)	1.44	2.82	5.43

Note. Self-administration elicits significantly more partners for all three time periods. Cell size is approximately 500. Significance tests were computed on logged data.

Source: Women's Health Study. From Table 22.2 (p. 440) in Tourangeau, R., & Smith, T. W. (1998). Collecting sensitive information with different modes of data collection. In M. Couper, R. Baker, J. Bethlehem, C. Clark, J. Martin, W. Nicholls, & J. O'Reilly (Eds.), *Computer-assisted survey information collection* (pp. 431–454). Copyright © 1998 by John Wiley & Sons, Inc. Reprinted by permission of authors and John Wiley & Sons, Inc.

computer-assisted self-interviewing (CASI), and computer-assisted personal interviewing (CAPI) to collect reports of the number of sex partners among an area probability sample in Cook County, Illinois. The results of this study are complex, involving three-way interactions, and the investigators present tabulated means only for the main effects. Table 7.3 shows the mean number of sex partners reported in each of three modes for each of three reference periods. The table combines responses of men and women because the mode-by-sex interaction was not significant, even though the expected pattern of men reporting fewer partners and women reporting more partners with self-administration was found (Tourangeau & Smith, 1996, p. 293). In this study, the mean number of partners reported was higher when ACASI was used, but not when CASI was used, and the effect was significant only for reports of the number of partners in the past 5 years and over the respondent's lifetime.

The ACASI experiment also manipulated question format and context to attempt to improve reports about threatening behaviors. In the question format manipulation, one version of the closed question about the number of sex partners presented high-frequency response categories and another version of the closed question presented low-frequency response categories. Each of these was followed by an open question that asked for an exact number of partners. This second response was compared with answers obtained to an open question administered as a third treatment. The main effects from this experiment are summarized in Table 7.4. The question format had a significant effect on the reports of the number of sex partners in the last year and last 5 years. The mean number of partners reported was highest with the high-frequency categories, lowest with the low-frequency categories, and intermediate with the open question. For the 5-year

TABLE 7.3
Mean Reported Sex Partners by Mode of Interview

Mode	*Past Year*	*Past 5 Years*	*Lifetime*
ACASI	2.26	4.52	7.05
CASI	1.87	2.99	5.75
CAPI	2.14	3.44	5.51

Note. The interview modes are audio-enhanced computer-assisted self-interviewing (ACASI), computer-assisted self-interviewing (CASI), and computer-assisted personal interviewing (CAPI). The effect of mode is significant for lifetime reports and marginally significant for 5-year reports. ACASI yields higher reports than CASI for lifetime and 5-year reports. There are higher order interactions involving mode of data collection. Cell sizes range from 86 to 104. Significance tests are based on logged data.

Source: ACASI Experiment. From Table 3 (p. 292) in Tourangeau and Smith (1996). Copyright © 1996 by the American Association for Public Opinion Research. Reprinted by permission of the University of Chicago Press and the author.

TABLE 7.4
Mean Reported Sex Partners by Question Format

Question Format	*Past Year*	*Past 5 Years*	*Lifetime*
Open	1.65	3.12	7.20
Closed-low	1.43	2.62	4.73
Closed-high	3.38	5.33	6.28

Note. The closed-low question format presented categories that implied a low frequency range; the closed-high question format presented categories that implied a high frequency range. The effect of question format is significant for the past year and past 5 years, and there is a significant interaction involving question format and mode of data collection. Significance tests are based on logged data.

Source: ACASI Experiment. From Table 3 (p. 292) in Tourangeau and Smith (1996). Copyright © 1996 by the American Association for Public Opinion Research. Reprinted by permission of the University of Chicago Press and the author.

and lifetime reports, there was an interaction between question format and mode that suggests that the effect of item format is reduced in the CAPI mode.

A third manipulation in the ACASI experiment placed questions that expressed either restrictive or permissive attitudes about sexual attitudes before the questions about the number of sex partners. The main effects reported in Table 7.5 suggest that the number of partners reported is higher in the restrictive context, but the difference is significant only for reports about partners in the last 5 years. There is a three-way interaction involving sex of the respondent, mode of asking the question, and context, such that the average number of sex partners reported by men and women is similar when a restrictive context and self-administration are used. Although such a complex three-way interaction requires replication, the results of these studies indicate that these methods can affect reporting of sex partners.

In all of these experiments about the reporting of sex partners, no external criterion is available, but a criterion of internal consistency in a closed

TABLE 7.5
Mean Reported Sex Partners by Question Context

Context	*Past Year*	*Past 5 Years*	*Lifetime*
Permissive	1.88	2.95	5.77
Restrictive	2.28	4.24	6.34

Note. The permissive context preceded the question about sex partners with attitude items expressing permissive attitudes toward sexual behavior; the restrictive context presented items expressing restrictive attitudes. The effect of question context is significant for the past 5 years but is qualified by higher order interactions. Significance tests are based on logged data.

Source: ACASI Experiment. From Table 3 (p. 292) in Tourangeau and Smith (1996). Copyright © 1996 by the American Association for Public Opinion Research. Reprinted by permission of the University of Chicago Press and the author.

population is used to diagnose the presence of response error. Evidence about men's and women's attitudes toward sexual behavior provide a basis for speculations about the different probable direction of reporting error for men and women (e.g., Smith, 1992, p. 320). But because there is no criterion, we cannot say which experimental manipulations produce more accurate reports in any absolute sense, although we can identify which manipulations produce reports that seem to be consistent with accurate reports. That is, we tend to interpret manipulations that reduce the number of partners reported by men or increase the number of partners reported by women as increasing the accuracy of self-reports.

REPORTING OF ABORTION

The second example to be considered here is the reporting of abortion. A comparison between responses to the National Survey of Family Growth (NSFG) and data compiled from abortion providers by the Alan Guttmacher Institute (AGI) illustrates the use of aggregate comparisons to diagnose the existence of a problem with self-reports and the direction of overall error. Jones and Forrest (1992) used the detailed retrospective fertility histories in the NSFG that cover the 3 or 4 years preceding the interview to estimate the number of abortions performed in the United States based on the NSFG. They compute these estimates for 11 years covered by the fertility histories in the 1976, 1982, and 1988 NSFG surveys. They then compare the survey-based estimates to the estimates of the number of abortions performed in each year that the AGI compiles from abortion providers. Their analysis suggests that, for example, 45% of the abortions actually performed in the 3 years covered by the 1976 NSFG were reported in the survey. (See Table 7.6, which shows the proportion of the abortions estimated from AGI data that are reported for the period covered by each of the three NSFG surveys.) In each of the 11 years for which Jones and Forrest used the NSFG fertility histories to make this comparison, the NSFG yields a substantially lower estimate of the number of abortions performed in that year than the AGI

TABLE 7.6
Percentage of Abortions Estimated as Actually Having Occurred in the United States as Reported in the 1976, 1982, and 1988 Cycles of the NSFG

	Percentage Reported
Cycle II, 1976 (1973–1975)	45
Cycle III, 1982 (1979–1982)	48
Cycle IV, 1988 (1984–1987)	35

Note. Source: National Survey of Family Growth. From Table 1 in Jones, E. F., & Forrest, J. D. (1992). Underreporting of abortion in surveys of U.S. women: 1976 to 1988. *Demography, 29*, 113–126. Reprinted by permission of the Population Association of America and the authors.

data. In reviewing these data, Jones and Forrest concluded that "... neither the incidence nor the trend in the number of abortions [using the AGI data as the criterion] can be inferred from the NSFG data" (p. 117), and they found similar deficiencies with the National Surveys of Young Women and the National Longitudinal Surveys of Work Experience of Youth (NLSY). (They also note that confidential questions in the 1988 NSFG and NLSY show promise for increasing the number of abortions reported.)

A portion of the sample for the Women's Health Study was selected from two Chicago clinics so that reports about abortion experiences could be validated by comparison with clinic records. This study illustrates the reverse record-check methodology, in which a sample is drawn from a source that can provide records to use in validating responses. The study attempted to improve the reporting of abortion by varying several factors: whether questions were self-administered or interviewer-administered, whether the medium for recording answers was paper or computer, whether the interview took place in the woman's home or at a neutral site, whether nurses or regular field staff conducted the interview, and whether questions about abortion were initially asked as part of a series of questions about medical procedures or in a series of questions about pregnancies and their outcomes. None of these experimental variables affected reports of having had at least one abortion. About 74% of the women in the clinic sample reported having ever had an abortion, and only 52% reported an abortion at roughly the time recorded in the clinic records (Tourangeau, Rasinski, Jobe, Smith, & Pratt, 1997, p. 359).

An experimental pretest for Cycle V of the NSFG that was conducted in 1993 tested methods that might improve the reporting of abortion (Lessler, Weeks, & O'Reilly, 1994). The pretest used a sample of women in three pairs of matched sites who had participated in the 1991 National Health Interview Survey. Results from the NSFG pretest, shown in Table 7.7, unlike those just reported for the Women's Health Study, which used a different sampling strategy, show some increases in the level of reporting. The study manipulated features of the interview and incentives. Women were assigned to one

TABLE 7.7
Percentage of Women Reporting They Ever Had an Abortion by Site, Incentive, and Mode

	In-Home (No Money)	*In-Home ($20)*	*Neutral Site ($40)*
No ACASI	14	22	29
ACASI	25	30	Not applicable

Note. ACASI administration followed interviewer administration. Both the neutral site-$40 incentive and ACASI increase the number of women reporting they ever had an abortion.

Source: NSFG Cycle V Pretest. From Exhibit C in Lessler, Weeks, and O'Reilly (1994). Reprinted by permission of the authors.

of three interview treatments: CAPI interview in the home, CAPI in the home with a short ACASI supplement, or CAPI interview at a neutral site. For the incentive manipulation, $20 was offered to all in-home respondents in three of the sites and no incentive was provided to their in-home counterparts in the matched sites; all respondents interviewed in a neutral site were paid $40 and reimbursed some expenses. The results of the experiment suggest that interviewing in a neutral site (with a $40 incentive) and using ACASI both similarly increase the proportion of women who report that they ever had an abortion.

REPORTING OF MARIJUANA USE

The final example concerns reporting of marijuana use. Unlike the previous two behaviors, for marijuana use there are neither records from providers or a closed population comparison to use in evaluating estimates or identifying, even experimentally, the direction of reporting error. Instead, reasoning by analogy with other behaviors for which records or other criteria are available and knowing that marijuana use is illegal, researchers make the assumption that marijuana use is underreported and that methods that increase reporting improve reporting.

The 1990 National Household Survey of Drug Abuse (NHSDA) Field Test drew on extensive preliminary investigations to test methods that might improve the reporting of drug use. The study used a multistage probability sample of households in 33 metropolitan areas. The experimental treatments included the 1990 NHSDA questionnaire and a new instrument crossed with two modes of administration, self-administered questionnaire (SAQ) and interviewer-administered paper-and-pencil interview (PAPI). The new instrument revised the 1990 NHSDA questions in an attempt to make them more internally consistent and to remove ambiguities. Results from these experiments are illustrated by the result in Table 7.8. These results show that the proportion of the sample reporting marijuana use is higher when questions are self-administered than when they are administered by an interviewer. The new question wording did not have a significant effect.

These examples illustrate that reports about very different types of threatening behaviors can be affected—and possibly improved—by manipulating the mode of interview and other features of the interview. Although the use of a self-administered instrument does not affect reporting in the direction we think of as improving reporting in every experiment in which it has been used, it has been evaluated frequently, and it often seems to improve reporting of threatening behaviors. Moreover, the effect is often greatest for reports about recent behavior, which are arguably more threatening. (See Tourangeau & Smith, 1998, for a summary).

TABLE 7.8
Estimates of Prevalence of Reported Marijuana Use (Percentage) by Questionnaire Wording and Mode of Administration

	By Question Wording		*By Mode*	
	Current	*New (Clarified)*	*SAQ*	*PAPI*
Lifetime	35.47	36.18	36.71	34.96
Past year	7.72	7.53	8.64	6.63
Past 30 days	3.94	4.16	5.02	3.11

Note. Effect of mode is significant for past 30 days and past year. Effect of question wording is not significant. Effect is clearest for most recent use.

Source: 1990 National Household Survey of Drug Abuse Field Test. From Table 7-1 in Turner, Lessler, and Devore (1992). Reprinted by permission of the author.

Two important questions raised by these experiments are (a) the extent to which the experimental treatments actually address sources of error associated with the sensitivity of the topics as opposed to other sources of reporting error, and (b) whether there are other sources of error due to the sensitivity of the topic that remain to be addressed. I consider next some issues suggested by these questions.

A MODEL OF MODE EFFECTS

Tourangeau and Smith (1996) have developed a model of mode effects that summarizes how self-administration and the use of computer-assisted data collection might affect the accuracy of self-reports. The model suggests that self-administration is effective because it increases the privacy of reporting, and the use of computer-assisted modes is effective because it enhances the perceived legitimacy of the data collection effort.

Their interpretation of the effect of privacy is supported by experiments that apply utility theory to studying threatening questions. When used to study answers to threatening questions, utility theory is concerned with how a respondent's risk-taking behavior (reporting disapproved facts about oneself) is affected by their perception of what is to be lost or gained by reporting (see Nathan, Sirken, Willis, & Esposito, 1990). In these experiments, subjects were presented with hypothetical scenarios describing a respondent in an interview. The subject rated the likelihood that the respondent would answer truthfully about a specific threatening topic in the situation described in the scenario. As the results in Table 7.9 indicate, the rated likelihood of a truthful response is greater in the more private situation; that is, when the respondent's family is not at home, at least when the interviewer is in her 50s. Moreover, other ratings by subjects (not shown in

TABLE 7.9
Effects of Interviewer Age and Privacy on Likelihood of Truthful Response

	Interviewer Age		
	20s	*50s*	*Overall*
Privacy			
Family not home	6.19	7.81	7.00
Family home	6.50	5.40	5.96

Note. Higher number means a greater likelihood of a truthful response. Main effect of privacy and privacy by interviewer age interaction are significant.

Source: Rasinski, K. A., Baldwin, A. K., Willis, G. B., & Jobe, J. B. (1994). Risk and loss perceptions associated with survey reporting of sensitive behaviors. In *Proceedings of the Section on Survey Research Methods of the American Statistical Association* (pp. 497–502). Washington, DC: The American Statistical Association. Reprinted by permission of the author.

Table 7.9) indicated that the perceived risk of disclosure to one's spouse and of embarrassment in front of the interviewer affected how likely the subjects thought it was that the respondent would tell the truth for a more threatening and a less threatening survey topic, respectively.

These vignette studies, and the results from the experiments reviewed earlier suggest that respondents are more comfortable revealing socially threatening things about themselves when they can do so privately. Thus, the process of reporting about sensitive topics is affected by situational variables. It is tempting to conclude that the response processes that are affected by manipulating the mode of the interview are conscious processes, but even the vignette studies, which simulate conscious decision making, do not demonstrate that the actual behavior of respondents during an interview is guided by conscious calculation. And it is quite possible that it is not.

WHAT MAKES A QUESTION THREATENING

With these examples as background, I would like to consider what directions the research just reviewed suggests for future research about reports of threatening behaviors. To do this, I first consider where errors in self-reports about threatening topics might originate and then what makes a question threatening.

Current views of errors in reports of threatening questions suggest four origins for reporting errors. Three of these—cognitive processing, the social organization of the behavior being reported about, and task requirements—affect all reports of behaviors. But the pattern of overreporting of desirable behaviors and underreporting of undesirable behaviors that is often

found when there is a criterion available to determine the direction of reporting error suggest that there are also sources of error specific to threatening topics.

The application of information-processing models to the question–answering process suggests that errors may originate at the stages of comprehension and interpretation of the question, retrieval, evaluation of the candidate answer, and reporting. Similarly, the way events are organized in respondents' lives may make them more or less difficult to report about accurately, particularly given the estimation methods and heuristics that respondents use to construct self-reports. Behaviors are organized in socially patterned ways in respondents' lives, and this affects how important the behavior is to the respondent, how frequently and regularly the behavior is engaged in, how similar each instance of the behavior is to other instances, how much social reinforcement the behavior receives, and how distinct it is from other, similar, behaviors. Because these features of a behavior interact with the estimation methods respondents use to affect how easy it is to report accurately about the behavior, accuracy and inaccuracy in reporting are also socially patterned. Thus, information about a class of events that is salient to the respondent and simple in its structure may be easier to report about accurately than other classes of events or behaviors.

Table 7.10 speculates that the three threatening topics I have reviewed probably differ in how they are organized in the experience of respondents (see Schaeffer, 1994). For example, sex partners are probably not acquired nor abortions undergone at regular intervals, but the use of marijuana may be regularly patterned for some respondents. The speculations in Table 7.10 are a reminder that the patterning and organization of threatening behaviors varies across respondents (and across groups of respondents) who engage

TABLE 7.10
Speculation on Social Organization of Three Threatening Behaviors

	Sex Partners	*Abortion*	*Drug Use*
Importance of category to respondent	variable	probably often high	variable, probably increases with frequency
Frequency	low for most, high for a few	.low for most, high for a few	low for most, high for a few
Regularity	low	low	variable
Similarity	probably increases with number	low for most	increases with frequency and regularity
Social reinforcement	low to moderate	low	may be high among social group
Clarity (distinct from similar events)	generally high	generally high	may be reduced if multiple drugs used

in such behaviors. Reports of such behaviors are subject to the same memory errors (such as omissions) as are reports of more mundane activities. But in the case of threatening behaviors, even errors of memory and processing may be biased toward a conventional, socially nonthreatening answer. That is, the nonthreatening answer to a threatening question—and an important source of response bias—may be incorporated in the heuristics that respondents use to construct their answers during an interview.

Consider sources of response error that are associated with the level of threat presented by the survey question. To be comprehensive, a theory of question threat would have to encompass topics as diverse as income, voting, intravenous drug use, and masturbation, and I have not found such a fully developed theory in the literature. But to begin, we can distinguish, as others have, between threatening questions and threatening answers. Threatening questions pose what Lee and Renzetti (1990) called an intrusive threat. Their discussion suggests that respondents may perceive a question as an intrusive threat if they consider the topic private or personal, if the question concerns deviant acts, if the topic is of interest to those in power, or if it is sacred to the respondent. Thus, respondents may consider a question threatening regardless of their answer. For example, they may consider their income to be "nobody's business" in general, no matter what their income, even if they might discuss it frankly under certain circumstances.

Most studies have been concerned with response situations in which a respondent perceives a question as threatening because of the respondent's (true) answer. A respondent who has never used intravenous drugs may recognize the question as intrusive, but answer "no" to the question without objecting. If the threat of a question varies depending on the answer, then the risks associated with admitting threatening behaviors depend on the respondent's behavior in that specific behavioral domain. A comprehensive list of the risks involved in giving a threatening answer, such as that begun in Table 7.11, would include the risks associated with having the respondent's answer disclosed to others outside the interview situation. These risks include the risk that others would be in a position to do any of the following: have information that would help them obtain access to the respondent, know that the respondent has something that the others want, know that the respondent has engaged in illegal behaviors, know that the respondent has engaged in behaviors counter to values that the others (or some third party) might hold, make other negative inferences about the respondent, or learn that the respondent has kept the information secret. Even if respondents were not concerned about the risks of disclosure, their answers might reflect the pressures of answering to an interviewer in a social situation. Thus, respondents might take into account (although not necessarily consciously) that an answer might reveal that they engaged in behaviors that are counter to values that the interviewer might hold, or that

TABLE 7.11
Risks and Losses in Answering Truthfully

Risk	*Loss*
Disclosure to Outsiders	
Improved access to *R*	Further intrusions
R has something others want	Further solicitations
Illegal behavior	Embarrassment, punishment
Counter (others') values	Embarrassment
Negative inferences	Embarrassment
Reveal past secrecy	Embarrassment, violation of openness
Disclosure to Interviewer	
Counter interviewer's (possible) values	Disagreement, embarrassment
Negative inferences by interviewer	Embarrassment
Painful feelings	Remember painful feelings
Counter (own) values	Threat to self

the interviewer might make negative inferences about them based on an answer. Finally, giving a threatening answer might raise painful or stressful feelings or require that respondents admit to having done things that they wish they had not.

Each of these risks has threatened losses associated with it. These losses include further intrusions or solicitations, embarrassment, painful memories, and threat to the respondent's self-concept. Experimental manipulations that increase the legitimacy of the survey or reduce interaction with the interviewer presumably reduce these perceived risks of disclosure and embarrassment and also any lack of frankness that might result from these sources. In contrast, threats to the respondent's self-concept have not been addressed by the experimental manipulations described earlier. A possible exception is the restrictive–permissive context manipulation in the reporting of sex partners, which may work, in part, by making specific personal values salient to the respondent when they construct their answers.

It is not obvious exactly how the mode of interview or other such manipulations might affect cognitive processing. Manipulations such as the use of a self-administered response mode or computerized instruments certainly could affect respondents' conscious calculations of risks and their likelihood and thereby influence a respondent's decision to report. This possibility is compatible with the view that overreporting of desirable behaviors and underreporting of undesirable behaviors is intentional or deliberate. Using the terms of the information-processing model described earlier, this might suggest that when a respondent retrieves a sensitive answer, he or she then decides whether or not to modify it before reporting. Manipulations such as the use of self-administered or computerized instruments may reduce such errors due to the sensitivity of a topic that occur in the final stages of processing.

But whereas such conscious calculations may be responsible for some reporting errors due to the sensitivity of the topic, and for part of the effect of interview mode described earlier, other processes may be relevant for some respondents or some behaviors. I have already suggested that nonthreatening answers may be incorporated into the heuristics and estimation methods that respondents use to construct survey answers. For example, a nonthreatening value may be retrieved and used as an anchor in an anchor-and-adjust response strategy.

And there are other related possibilities. For example, as my epigraph indicates, ethicists have argued about whether a falsehood told to someone who has no right to the requested information is a lie. Regardless of the moral status of such falsehoods, they have a routine social function in smoothing casual interactions and deflecting rude inquiries of all kinds: If someone asks a question that is none of their business, the recipient of that question may prefer a routine denial to the dispreferred (and possibly implicating) act of refusing to answer. Furthermore (and partly because of their conventional nature), nonthreatening answers may be embedded in the public biographies—personal histories available for public consumption—that people construct about themselves. For some people, parts of these public biographies may, over time, become incorporated into a public self-concept or may replace actual history in memory. Thus, a denial that one has used marijuana may be produced almost automatically because it is the conventional answer to a question that is nobody's business or because it is part of a public self. As such, the denial requires no extensive retrieval or active judgment about the risks of reporting in an interview—it is a fast and safe solution to the task of answering a threatening question. And with repetition, the public, socially acceptable answer may replace other historical answers entirely.

We can find some insight into automatic processing and the role of cognitive structures in such processing in research on racial stereotypes, a kind of cognitive structure (Devine, 1989). This research suggests that relevant cues from the environment may automatically evoke stereotypes that are used in cognitive processing. Processing and decision making will draw on these automatic structures unless competing values and beliefs are made conscious. When competing values and beliefs are made conscious, they may then guide actual behavior.

In the survey interview, automatic processing could be activated by cues that a question presents and could affect the reporting of answers to threatening questions in at least two ways. First, a question may activate what Dykema (1996) called a personal-event schema. This cognitive structure

incorporates the individual's definition of the behavior, information about its frequency, as well as links to affect and the self-concept. We might also think of this schema as having several layers that distinguish information about the individual's behavior in a public biography, which can be readily reported, from information that is to be kept more private. Second, a question about a threatening topic might evoke interactional conventions for dealing with intrusive questions. These conventions could also affect answers to a self-administered form, as Schwarz (1990) argued. I would speculate that recipients of intrusive questions routinely use denial as an automatic, socially routine, defensive strategy. Such conventional denials could be offered simply because an intrusive question was asked and without any calculation of risks. In other words, respondents may deny engaging in threatening behaviors partly because they routinely do so.

Automatic processing of this type could affect several stages of cognitive processing—for example, how the question or the cognitive task is understood, retrieval, or evaluation of a candidate answer (see Table 7.12). Respondents might interpret threatening questions in a way that makes the behavior being asked about different from the behaviors they themselves engage in and thereby distance themselves from the behavior. Respondents may also treat the cognitive task posed by the question as one of classifying themselves as more like someone who engages in the behavior or more like someone who does not. For example, the pragmatic meaning of a question about abortion that a respondent constructs may be to determine whether the respondent is more like someone who has an abortion or more like someone who does not. If some respondents treat threatening questions in this way, there is an opportunity for other cognitive structures (e.g., a stereotype of what someone who has an abortion is "like") to play a role in the response process. In either of these two cases, the respondent's cogni-

TABLE 7.12
Examples of Possible Automatic Processing in Reports of Sensitive (Threatening) Behaviors

Stage	*Example—Threat to Self*
Comprehension of question	(Ambiguous) terms interpreted in ways that distance *R* from threatening events or behaviors
Interpretation of cognitive task	Question is asking *R* to classify self as more like someone who engages in behavior or more like someone who does not
	Question is "nobody's business," further processing is not required, standard social answer may be given
Retrieval	Full search not undertaken because question "does not apply" to *R*
Judgment	Standard socially acceptable answer is adequate

tive processing may be superficial because once the question has been redefined, it no longer requires a threatening answer.

CONCLUSION

Improving the accuracy of answers to questions on threatening topics has proved particularly challenging and somewhat daunting. As the review with which I began this chapter suggests, recent research appears to have made some progress in reducing errors due to the sensitivity of a topic, principally by increasing the privacy of responding. However, as this more speculative discussion suggests, further improvements are likely to require more specific theorizing about the stages of processing at which errors arise and about the role of automatic processing and conscious decision making in that processing. In addition, we must probably consider that such response errors may have different origins for different respondents and different topics and thus require multiple solutions.

REFERENCES

Bok, S. (1978). *Lying: Moral choice in public and private life*. New York: Pantheon.

Devine, P. G. (1989). Stereotypes and prejudice: Their automatic and controlled components. *Journal of Personal and Social Psychology, 56*, 5–18.

Dykema, J. (1996). *Events, instruments, and reporting errors: Combining knowledge from multiple perspectives*. Unpublished Master's Thesis, University of Wisconsin–Madison.

Jones, E. F., & Forrest, J. D. (1992). Underreporting of abortion in surveys of U.S. women: 1976 to 1988. *Demography, 29*, 113–126.

Lee, R. M., & Renzetti, C. M. (1990). The problems of researching sensitive topics. *American Behavioral Scientist, 33*, 510–528.

Lessler, J. T., Weeks, M. F., & O'Reilly, J. M. (1994). Results from the National Survey of Family Growth Cycle V Pretest. In *Proceedings of the American Statistical Association, the Section on Survey Research Methods* (pp. 64–70). Washington, DC: American Statistical Association.

Locander, W. B., Sudman, S., & Bradburn, N. (1976). An investigation of interview method, threat and response distortion. *Journal of the American Statistical Association, 71*, 269–275.

Nathan, G., Sirken, M., Willis, G., & Esposito, J. (1990, November). *Laboratory experiments on the cognitive aspects of sensitive questions*. Paper presented at the International Conference on Measurement Errors in Surveys, Tuscon, Arizona.

Parry, H. J., & Crossley, H. M. (1950). Validity of responses to survey questions. *Public Opinion Quarterly, 14*, 61–80.

Rasinski, K. A., Baldwin, A. K., Willis, G. B., & Jobe, J. B. (1994). Risk and loss perceptions associated with survey reporting of sensitive behaviors. In *Proceedings of the Section on Survey Research Methods of the American Statistical Association* (pp. 497–502). Washington, DC: The American Statistical Association.

Schaeffer, N. C. (1994). Errors of experience: Response errors in reports about child support and their implications for questionnaire design. In N. Schwarz & S. Sudman (Eds.), *Autobio-*

graphical memory and the validity of retrospective reports (pp. 141–170). New York: Springer-Verlag.

Schwarz, N. (1990). Assessing frequency reports of mundane behaviors: Contributions of cognitive psychology to questionnaire construction. In C. Hendrick & M. S. Clark (Eds.), *Research methods in personality and sociology* (pp. 98–119). Newbury Park, England: Sage.

Smith, T. W. (1992). A methodological analysis of the sexual behavior questions on the General Social Surveys. *Journal of Official Statistics, 8*(3), 309–325.

Tourangeau, R., Rasinski, K., Jobe, J., Smith, T. W., & Pratt, W. F. (1997). Sources of error in a survey on sexual behavior. *Journal of Official Statistics, 13*, 341–365.

Tourangeau, R., & Smith, T. W. (1996). Asking sensitive questions: The impact of data collection mode, question format, and question context. *Public Opinion Quarterly, 60*, 275–304.

Tourangeau, R., & Smith, T. W. (1998). Collecting sensitive information with different modes of data collection. In M. Couper, R. Baker, J. Bethlehem, C. Clark, J. Martin, W. Nicholls, & J. O'Reilly (Eds.), *Computer-assisted survey information collection* (pp. 431–454). New York: Wiley.

Turner, C. F., Lessler, J. T., & Devore, J. (1992). Effects of Mode of Administration and Wording on Reporting of Drug Use. In C. F. Turner, J. T. Lessler, & J. C. Gfroerer (Eds.), *Survey measurement of drug use* (pp. 177–220). Rockville, MD: National Institute on Drug Abuse, U.S. Department of Health and Human Services.